

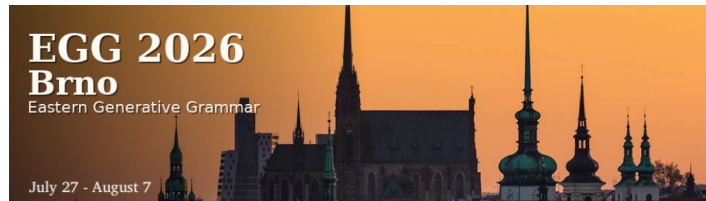


The Functional Architecture of Nominals

Kerstin Hoge

University of Oxford

<https://kxhoge.github.io/resources.html>



27 July – 31 July 2026

About this course

- *What does this course address?*

Since the 1980s, work in generative syntax has seen much discussion of the inventory and hierarchical ordering of functional categories in the nominal domain. This course provides an introduction and overview to these foundational developments in the generative analysis of nominal constructions. It aims to equip you with the analytical skills and the theoretical and empirical knowledge needed for an appreciation of the issues discussed and the ‘big picture’ questions, prominent amongst which are questions concerning the universality and cross-linguistic uniformity of the functional structure associated with the lexical core.

About this course (cont.)

- *What will I have learnt by the end of this week?*

Upon completion of the course, you will

- be familiar with the historical trajectory of the expansion of functional structure within nominal projections;
- have gained an understanding of the theoretical debate on the phrase structure of nominals;
- be able to evaluate arguments for and against postulating functional projections in the nominal domain;
- have acquired the theoretical background that is necessary to understand and situate current research on nominal syntax.

About this course (cont.)

- *How is this course taught?*

The course is taught in five lectures. You are welcome to ask clarification and follow-up questions during the lectures. Since syntax is best learnt hands-on, each session will feature a brief data-analysis segment, which you can tackle either independently or by collaborating with your seat neighbour.

Course outline

Monday	Approaches to the functional structure of nominals
Tuesday	The NP vs. DP debate
Wednesday	Pronouns
Thursday	Demonstratives
Friday	Diminutives

Approaches to the Functional Structure of Nominals

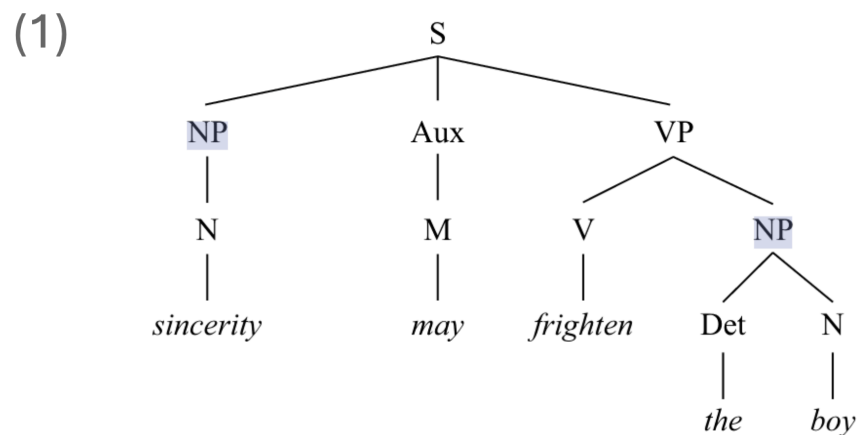
Outline

1. Approaches to nominal structure: The eras tour
2. Arguing for functional projections: Introducing DP
3. Functional projections between N and D: Number
4. Summary and outlook

Approaches to the Functional Structure of Nominals

Approaches to nominal structure: The eras tour

- In the history of generative grammar, nominal expressions have received various syntactic analyses.



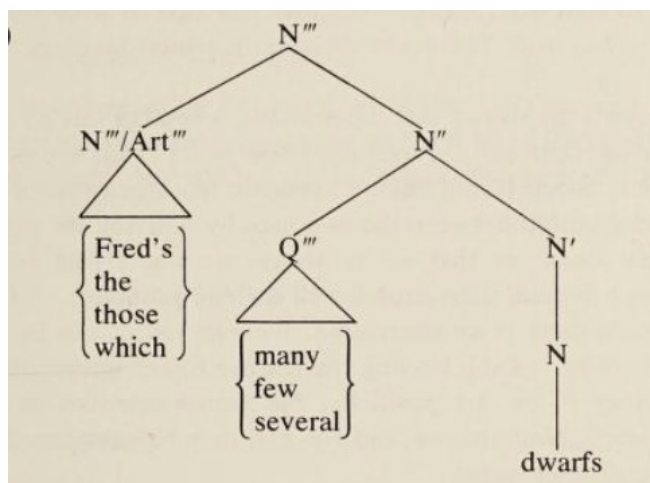
(Chomsky 1965: 69, ex. (3))

Approaches to the Functional Structure of Nominals

Approaches to nominal structure: The eras tour

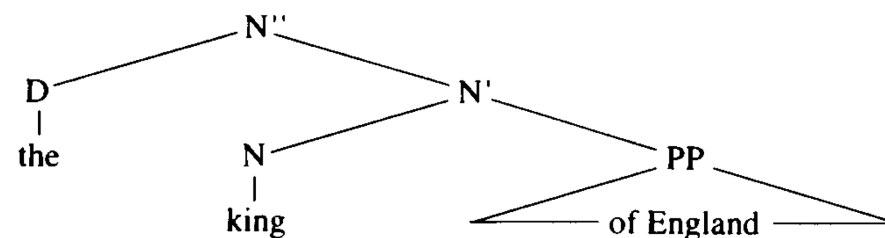
- With the advent of X-bar structure (Chomsky 1970, Jackendoff 1977), the flat structures of the Standard Theory (Chomsky 1965) gave way to representations in which determiners occurred as the specifier rather than the sister of N.

(2)



(Jackendoff 1970: 105, ex. (5.5))

(3)



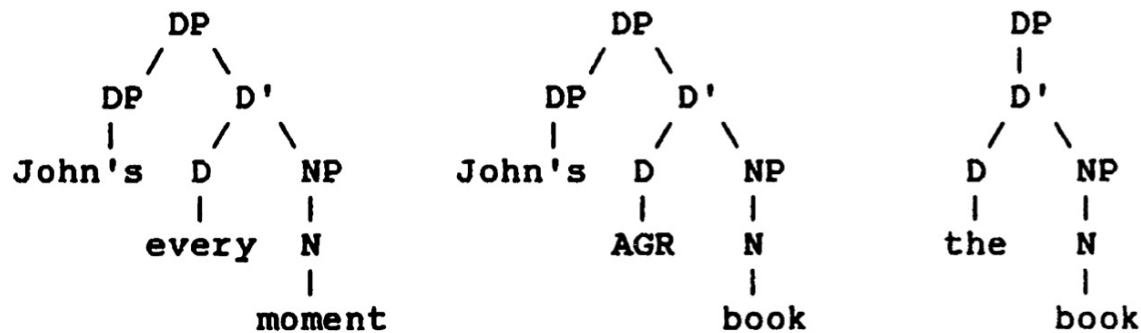
(Radford 1988: 173, ex. (24))

Approaches to the Functional Structure of Nominals

Approaches to nominal structure: The eras tour

- Following the reanalysis of S and S' in X'-theoretical terms (Chomsky 1981, 1986), Abney (1987) develops the DP-analysis of nominals (cf. earlier proposals by Brame 1981, 1982 and Szabolsci 1981 among others, see Abney 1987: 77).

(4)

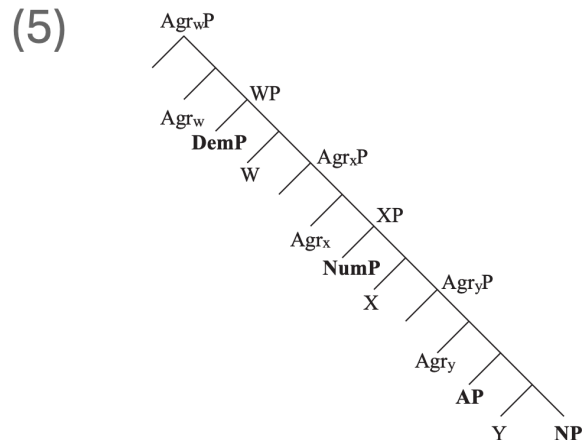


Abney 1987: 25, ex. (16))

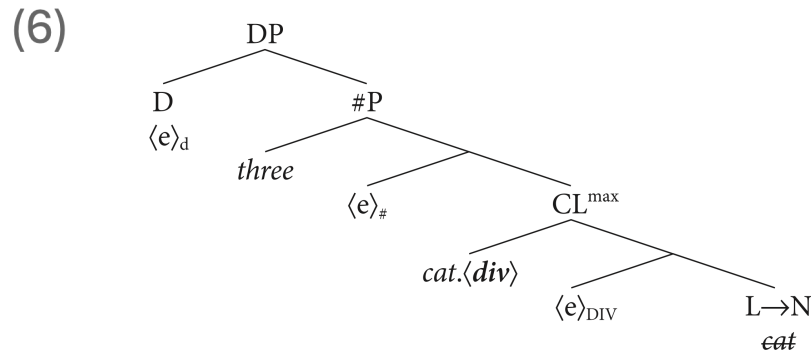
Approaches to the Functional Structure of Nominals

Approaches to nominal structure: The eras tour

- The years following Abney's DP-hypothesis saw an expansion of functional categories in both the verbal (cf. Pollock 1989) and nominal domains, which meshed with and contributed to the development of cartographic (Rizzi 1997; Cinque 1999) and neo-constructionist (Halle & Marantz 1993; Marantz 1997; Borer 2003, 2005a, 2005b; Caha 2009; Starke 2009) approaches to syntax.



(Cinque 2020: 10, ex. (12))



(Borer 2005a: 109, ex. (27))

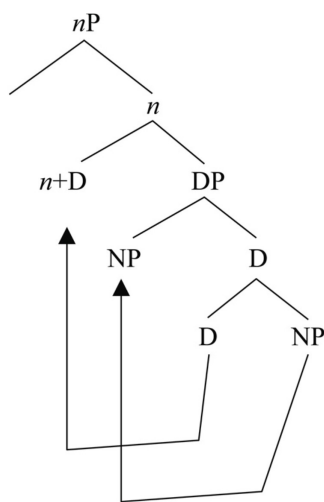
Approaches to the Functional Structure of Nominals

Approaches to nominal structure: The eras tour

- Yet there also exist proposals to reduce the functional structure of nominals and once again represent the noun as the syntactic core of a nominal expression:

‘...the head is now n^* (analogous to v^*) with the complement $[X (YP)]$. In this case $X = D$. D inherits the features of n^* , so YP raises to its SPEC, and D raises to n^* , exactly parallel to v^*P . Therefore, the structure is a nominal phrase headed by n^* , not a determiner phrase headed by D , which is what we intuitively always wanted to say’ (Chomsky 2007: 25-26).

(7)



(Punske 2014: 106)

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- The variety of extant approaches to nominal phrase structure (which range from the flat structures of the Standard Theory to the richly articulated structures of Cartography) brings into view the question of **what motivates and justifies the postulation of a functional category**.
- Syntactic argumentation draws on three types of evidence for postulating functional categories:
 - semantic;
 - morphological;
 - distributional / syntactic.

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Semantic arguments:

‘Lexical categories may be taken to express certain concepts, but in the context of clauses, these ‘lexical’ concepts are associated with additional notions. The idea is then that these additional notions are encoded in functional heads that are associated with the lexical head in question ...

Constituents headed by nouns denote entities (persons, things) but they also contain information concerning reference ... Since such information is not an inherent part of the noun, it is proposed that there is a specialized head D to encode the referential status of the nominal projection’
(Alexiadou, Haegeman & Stavrou 2007: 23-24).

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Morphological arguments:

‘In many languages when lexical heads are inserted into a sentence they do not come ‘bare’, that is as mere stems. Rather, they are associated with inflectional morphology. Because morphology is variable (for instance the verb may be associated with a choice of tenses) it is not taken to be an intrinsic part of the lexical head as such, but rather it can be argued that the inflectional morphemes constitute functional heads in the extended projection of the lexical head’ (Alexiadou, Haegeman & Stavrou 2007: 24).

Nouns may inflect for definiteness, as in Norwegian (Sunnhordland dialect):

- (8) a. ein bi:l b. bi:l-**en** di:-n
 INDF.SG.M car car-DEF.SG.{M} your-SG.M
 ‘a car’ ‘your car’
 (Enger & Corbett 2012: 291, ex. (4))

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Distributional / syntactic arguments:

‘Two types of argumentation are relevant here. These were formulated by Taraldsen (1990). The first type of argument essentially relates to the distribution of heads and is based on three widely accepted axioms of the Government and Binding model: (i) a head can only move to a head position; (ii) every head X_0 is the head of maximal projection X_n ; and (iii) a moved constituent must c-command its trace (Taraldsen 1990: 85-86). The second type of argument relates more to the distribution of XPs and is based on the premise that every X_n dominates at most one specifier’ ...

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Distributional / syntactic arguments (cont.)

... evidence that N may occupy more than one position in the nominal constituent could lead us to assume N-movement and hence to postulate specific head positions as landing sites for N) ...

(9) a. il mio Gianni
the my Gianni
'my Gianni'

b. il Gianni mio
the Gianni my
'MY Gianni'

[Italian]

c. Gianni mio
Gianni my
'my Gianni'

d. *mio Gianni
my Gianni
'my Gianni'

(cf. Longobardi 1994: 623, ex. (28))

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Distributional / syntactic arguments (cont.)

... if we observe that DP-internally, maximal projections may occupy different positions this can motivate postulating specifier positions, and by implication it provides indirect evidence for functional projections' ... (Alexiadou, Haegeman & Stavrou 2007: 26-28).

[If] adverbial modifiers are the specifiers of specialized projections ... [and] adjectival modifiers in the DP are seen as the analogies of adverbial modifiers in the clause, then again the functional projections postulated for hosting adverbial adjuncts in the clause could be replicated in the extended projection of the noun where they would host adjectives' (Alexiadou, Haegeman & Stavrou 2007: 26-28).

- (10) a. Mary frequently viciously criticized John.
b. Mary's frequent vicious criticism of John
(Alexiadou, Haegeman & Stavrou 2007: 28, ex. (20) & (21)).

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Abney's (1987) own arguments for the DP-hypothesis involve
 - theory-internal arguments
(relating to X-bar theoretical requirements);
 - parallelism arguments
(based on the uniformity of clausal and nominal structure);
 - head movement arguments (suggesting a head position above N).

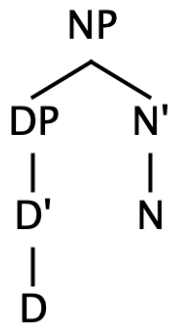
Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Theory-internal arguments: Projection

‘Interpreted strictly the X-bar requirement ... that non-head nodes be maximal projections, rules out the [standard NP] structure, inasmuch as DET is a non-head which is not a maximal projection. To preserve X-bar theory, we must modify the standard structure for the noun phrase to ...

(11)



Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

But the structure [in (11)] is made highly suspect by the fact that DetP (under this analysis) never contains any material except Det. It is difficult to motivate a phrasal node XP where there is no member of the class X which ever takes specifiers or complements. If it means anything to be a phrasal node, it is that the node in question dominates more than one word, at least potentially. This is the sense in which the standard analysis can be made to conform to the letter, but not the spirit, of X-bar theory' (Abney 1987: 288-289).

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Theory-internal arguments: Number of specifiers

Quantifiers and semi-numerals can appear in a position below D, preceding both N and any adjectival modifiers of N.

- (12) a. the **many** honest men
b. the **nearly a dozen** men who fell

‘If we wish to preserve the standard analysis, we must assume that these QP’s and NP’s do not occupy a lower spec position. One possibility would be that they are simply a species of prenominal adjective. There are a number of ways that they differ from descriptive adjectives, though, that render this hypothesis untenable’ (Abney 1987: 292).

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Descriptive noun phrases do not have the same distribution as APs:

- (13) a. the brave medical student
b. *the **nearly a doctor** medical student
(Abney 1987: 293, ex. (320))

- While the preferred order of prenominal adjectives can be manipulated for the sake of emphasis, ‘the requirement that QP and NP precede descriptive adjectives cannot be so readily violated’ (Abney 1987: 293):

- (14) a. a fancy new car
b. a NEW fancy car

- (15) a. the **many** honest men
b. *the HONEST **many** men
(Abney 1987: 293, ex. (321))

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Parallelism arguments: Parallel argument structure

‘Certain noun phrases – namely, derived nominals – [are] similar to sentences in their internal structure’ (Abney 1987: 32), viz. both nominals and clauses admit **internal** and **external** arguments (Lees 1960).

- (16) a. **Nero** destroyed **Rome**.
b. **Nero**’s destruction of **Rome**
(cf. Abney 1987: 32, ex. (19))

In both domains, the agent subject can bind a reflexive:

- (17) a. John portrayed himself.
b. John’s portrayal of himself
(Abney 1987: 33, ex. (20))

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Parallelism arguments: Identical agreement morphology

In Yupik, '[n]ouns – even concrete nouns – agree with their possessors. The agreement they show is the same agreement morpheme which is found on the verb, sharing even the same suppletions. Furthermore, the subject of the noun phrase takes ergative case, the case of subjects of transitive verbs' (Abney 1987: 39).

(18) a. angute-t kiputa-a-**t**
man-ERG.PL buy-OM.SG-SM.PL
'The men bought it.'

b. angute-k kiputa-a-**k**
man-ERG.DU buy-OM.SG-SM.DU
'The (two) men bought it.'
(cf. Abney 1987: 39, ex. (24))

(19) a. angute-t kuig-a-**t**
man-ERG.PL river-SG-AGR.PL
'the men's river'

b. angute-k kuig-a-**k**
man-ERG.DU river-SG-AGR.DU
'the (two) men's river'

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

- Head-movement arguments: Morphological mergers

Morphological mergers involve the combination of separate syntactic elements into a single form:

- (20) a. au < à + le [French]
 to + the
 b. everyone < every + one**
 (Abney 1987: 285, ex. (308))

‘If ... morphological mergers are made possible by head movement, we must assume that determiners are the heads of noun phrases’ (Abney 1987: 286-287).

Approaches to the Functional Structure of Nominals

Arguing for functional projections: Introducing DP

**Empirical evidence in support of the head-movement analysis of *someone* / *something* / *everyone* / *everything* is said to be provided by the acceptability contrast involving postnominal adjectives.

- (21) a. *a man good
b. Someone / something / ?everyone / everything good
(Abney 1987: 287, ex. (311))

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

- Ritter (1991) argues for the existence of a second nominal functional projection above N, Num(ber)P, on the basis of Hebrew free genitive constructions.
- In addition to free genitive constructions, Hebrew has so-called construct state constructions.

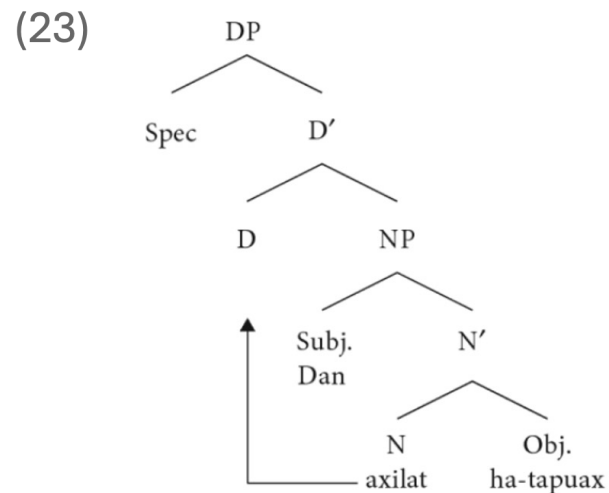
‘A construct state is a type of noun phrase containing a **bare genitive phrase** immediately following the head noun, that is a genitive phrase that is not overtly case marked’ (Ritter 1991: 38).

- (22) a. axilat **Dan** et ha-tapuax
eating Dan ACC the-apple
‘Dan’s eating of the apple’
b. *ha-axilat Dan et ha-tapuax
the-eating Dan ACC the-apple

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

- Assuming the possessor to be merged in the specifier of NP, the unacceptability of the definite article in the construct state follows straightforwardly if the construct state is derived by means of N-to-D movement of the possessed noun.



Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

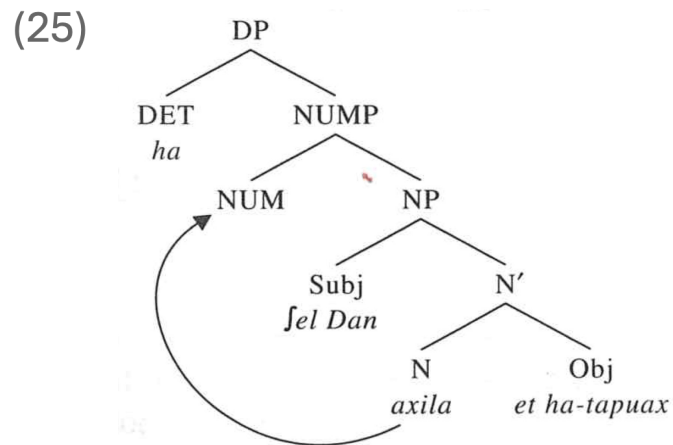
- In the free genitive construction, the possessed noun occurs with the definite article and the possessor is immediately preceded by an **overt genitive marker**.

(24) ha-axila **shel** Dan et ha-tapuax
 the-eating of Dan ACC the-apple
 ‘Dan’s eating of the apple’

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

- If the possessor and the possessed noun both originate within the NP, the free genitive construction, in which the possessed noun occurs between the definite article and the possessor, must be derived by raising N to 'another head position intermediate between DP and NP' (Ritter 1991: 43).

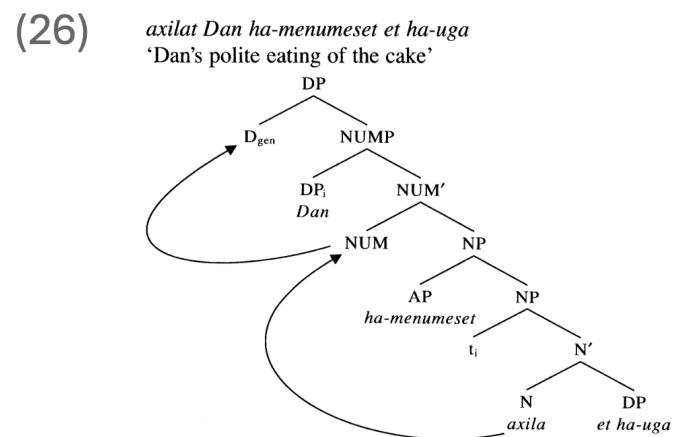


(Ritter 1991: 43, ex. (13))

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

- Considering evidence on the order of adjectives relative to the head noun and its argument in the construct state construction, Ritter argues for **all** (Hebrew) noun phrases to contain a NumP. In the construct state construction, both the possessed noun and the possessor move out of the NP in which they originate.



(Ritter 1991: 45, ex. (16))

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

Try your hand at some data analysis

- Hebrew *kol* ‘all’ occurs in the construct state but not in the free genitive construction.

- (27) a. kol ha-yeladim
all the-children / boys
‘all the children / boys’
b. *kol shel (ha-)yeladim
all of (the-)children / boys

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

Like other construct state genitives, *kol* construct state constructions

- cannot have a definite article in initial position:
 - (28) a. *ha-axilat Dan et ha-tapuax
the-eating Dan ACC the-apple
 - b. *ha-kol (ha-)yeladim
the-all (the-)children / boys
- and may occur with a pronominal clitic rather than a full bare genitive:
 - (29) a. axilat-o et ha-tapuax
eating-his ACC the apple
'his eating of the apple'
 - b. kul-am
all-3.M.PL
'all of them'

(

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

Note that *kol* ‘all’ cannot precede an indefinite plural NP or a definite singular NP:

- (30) a. **kol yeladim*
all children / boys
b. **kol ha-yeled*
all the-child /boy

Question 1: What is the syntactic category of *kol* ‘all’?

Question 2: What is the syntactic structure of *kol ha-yeladim* ‘all the children / boys’?

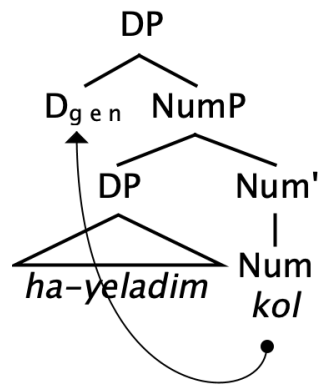
Question 3: What must you assume to be able to account for the absence of *kol* in the free genitive construction (**kol shel (ha-)yeladim*)?

Approaches to the Functional Structure of Nominals

Functional projections between N and D: Number

- In Ritter's analysis, quantifiers that occur in the quantifier construct state construction are 'NUMs that take no lexical complement, that is, no NP' (Ritter 1991: 55).

(31)



Ritter's analysis derives the fact that quantifiers like *kol* 'all' cannot occur in the free genitive construction by assuming that *shel* ('of')-insertion applies only to arguments inside NP.

Approaches to the Functional Structure of Nominals

Summary and outlook

- Following Abney's (1987) DP-hypothesis, much work in syntax has suggested and made use of an articulated functional structure inside nominal expressions. Ritter (1991) provides an early example of the proposal for a functional head other than D.
- Arguments made to corroborate the postulation of functional categories involve semantic, morphological and distributional / syntactic evidence. As illustrated in Abney's (1987) work, the arguments given may also be of a theory-internal nature.
- If the arguments for the DP-hypothesis rest at least in part on morphological evidence, what does this suggest for nominal structure in languages that lack overt definiteness marking, such as Czech?

(32) a. Kniha je na stole.

book is on table

'The book is on the table.'

b. Na stole je kniha.

on table is book

'There is a book on the table.'

Approaches to the Functional Structure of Nominals

References

- Abney, Steven. 1987. The English noun phrase in its sentential aspect. MIT PhD thesis.
- Alexiadou, Artemis, Liliane Haegeman & Melita Stavrou. 2007. *Noun phrase in the generative perspective* (Studies in Generative Grammar 71). Berlin & New York: Mouton de Gruyter.
- Borer, Hagit. 2003. Exo-skeletal vs. endo-skeletal explanations: Syntactic projections and the lexicon. In John Moore & Maria Polinsky (eds.), *The nature of explanation in linguistic theory*, 31–67. Stanford, CA: CSLI.
- Borer, Hagit. 2005a. *Structuring sense, vol. 2: The normal course of events*. Oxford: Oxford University Press.
- Borer, Hagit. 2005b. *Structuring sense, vol. 1: In name only*. Oxford: Oxford University Press.
- Brame, Michael. 1981. The general theory of binding and fusion. *Linguistic Analysis* 7, 277–325.
- Brame, Michael. 1982. The head-selector theory of lexical specifications and the nonexistence of coarse categories. *Linguistic Analysis* 10, 321–325.
- Caha, Pavel. 2009. The nanosyntax of case. University of Tromsø PhD thesis.
- Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 1970. Remarks on Nominalization. In Roderick A. Jacobs & Peter S. Rosenbaum (eds.), *Readings in English transformational grammar*, 184–221. Waltham, MA: Ginn.
- Chomsky, Noam. 1986. *Barriers*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2007. Approaching UG from below. In Uli Sauerland & Hans-Martin Gärtner (eds.), *Interfaces + recursion = language? Chomsky's Minimalism and the view from syntax–semantics*, 1–29. Berlin: Mouton de Gruyter.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. New York: Oxford University Press.
- Cinque, Guglielmo. 2020. *The Syntax of Relative Clauses: A Unified Analysis* (Cambridge Studies in Linguistics). Cambridge: Cambridge University Press. doi:[10.1017/9781108856195](https://doi.org/10.1017/9781108856195).
- Enger, Hans-Olav & Greville G. Corbett. 2012. Definiteness, Gender, and Hybrids: Evidence from Norwegian Dialects. *Journal of Germanic Linguistics* 24(4), 287–324. doi:[10.1017/S1470542712000098](https://doi.org/10.1017/S1470542712000098).
https://www.cambridge.org/core/product/identifier/S1470542712000098/type/journal_article.

Approaches to the Functional Structure of Nominals

References (cont.)

- Halle, Morris & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In Kenneth Hale & Samuel Jay Keyser (eds.), *The view from Building 20: Essays in linguistics in honour of Sylvain Bromberger*, 111–176. Cambridge, MA: MIT Press.
- Jackendoff, Ray. 1977. *X Syntax: A study of phrase structure*. Cambridge, MA: MIT Press.
- Longobardi, Giuseppe. 1994. Reference and proper names: A theory of N-Movement in syntax and Logical Form. *Linguistic Inquiry* 25, 609–665.
- Marantz, Alec. 1997. No escape from syntax: Don't try morphological analysis in the privacy of your own lexicon. *University of Pennsylvania Working Papers in Linguistics* 4(2), 201–225.
- Pollock, Jean-Yves. 1989. Verb movement, Universal Grammar, and the structure of IP. *Linguistic Inquiry* 20(3), 365–424.
- Punske, Jeffrey. 2014. Functional structure inside nominal phrases. In Andrew Carnie, Yosuke Sato & Daniel Siddiqi (eds.), *The Routledge handbook of syntax*, 97–120. Abingdon & New York: Routledge.
- Radford, Andrew. 1988. *Transformational grammar: A first course*. Cambridge: Cambridge University Press.
- Ritter, Elizabeth. 1991. Two functional categories in noun phrases: Evidence from Modern Hebrew. *Syntax and Semantics* 25, 37–62.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. In Liliane Haegeman (ed.), *Elements of grammar*, 281–337. Dordrecht: Kluwer Academic Publishers.
- Starke, Michal. 2009. Nanosyntax: A short primer to a new approach to language. *Nordlyd* 36(1), 1–6. doi:10.7557/12.213. <https://www.ub.uit.no/baser/septentrio/index.php/nordlyd/article/view/213>.
- Szabolcsi, Anna. 1981. The possessive construction in Hungarian: A configurational category in a non-configurational language. *Acta Linguistica Academiae Scientiarum Hungaricae* 31, 261–289.
- Taraldsen, Knut Tarald, Joan Mascaró & Marina Nespor. 1990. D-projections and N-projections in Norwegian. *Grammar in progress: GLOW essays for Henk van Riemsdijk*, 419–432. Dordrecht: Foris.